

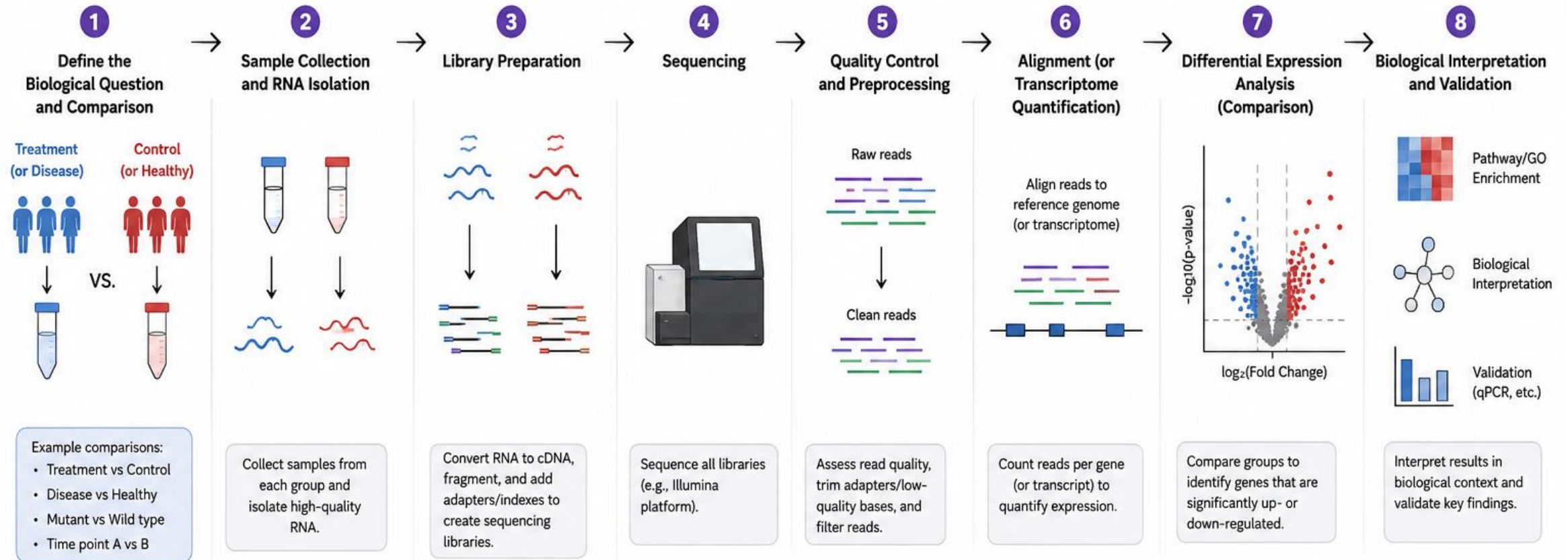
Differential Expression/Pathway Analysis

RNASeq data analysis: functional annotation

RNA-seq is always a **COMPARISON**



We compare groups to identify genes/transcripts that are differentially expressed.



Key Takeaway

RNA-seq is not just about measuring expression — it's about **COMPARING** groups.
The power of RNA-seq comes from discovering what changes between conditions.

Treatment
(or Disease)

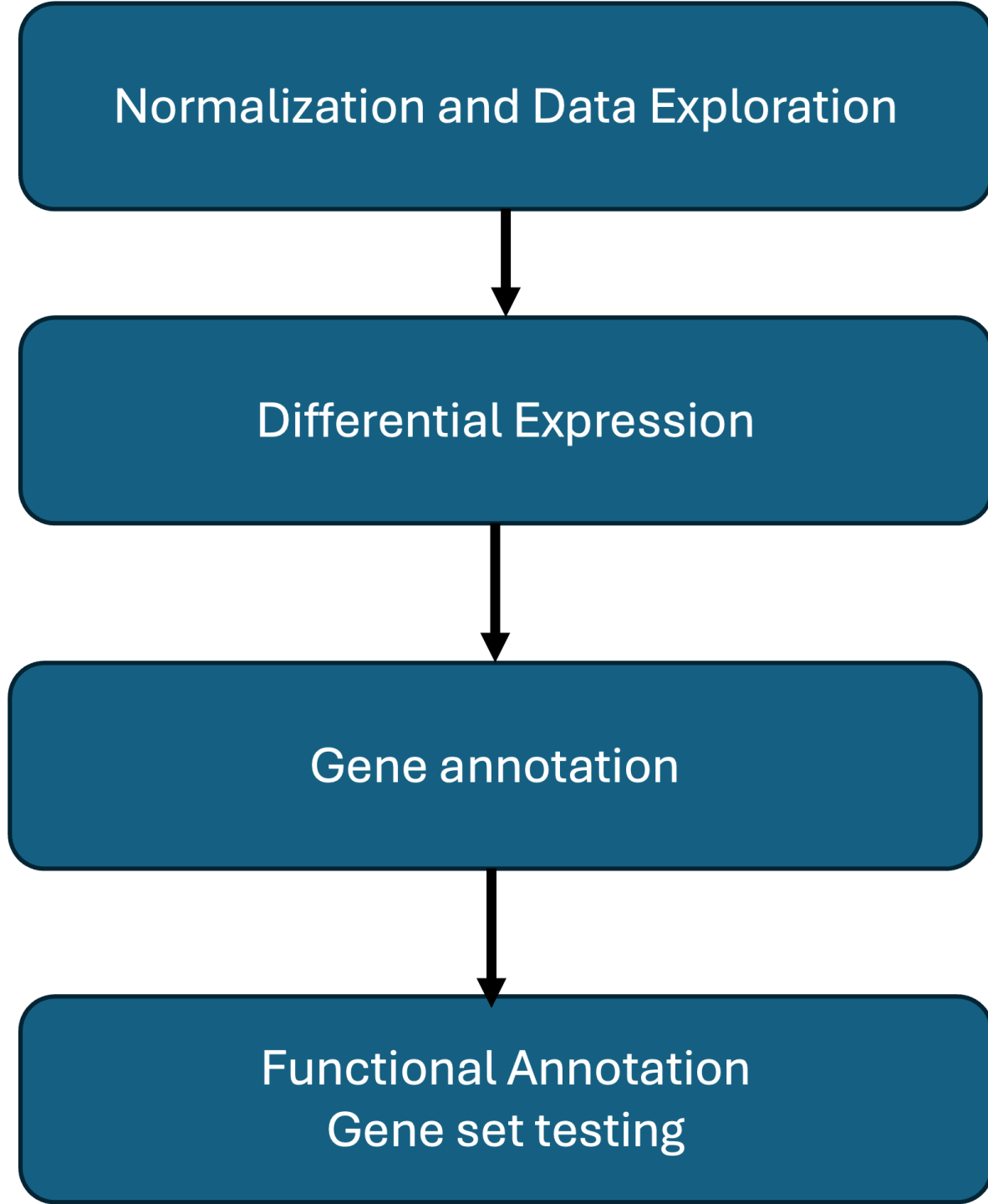


VS.

Control
(or Healthy)



RNAseq analysis workflow

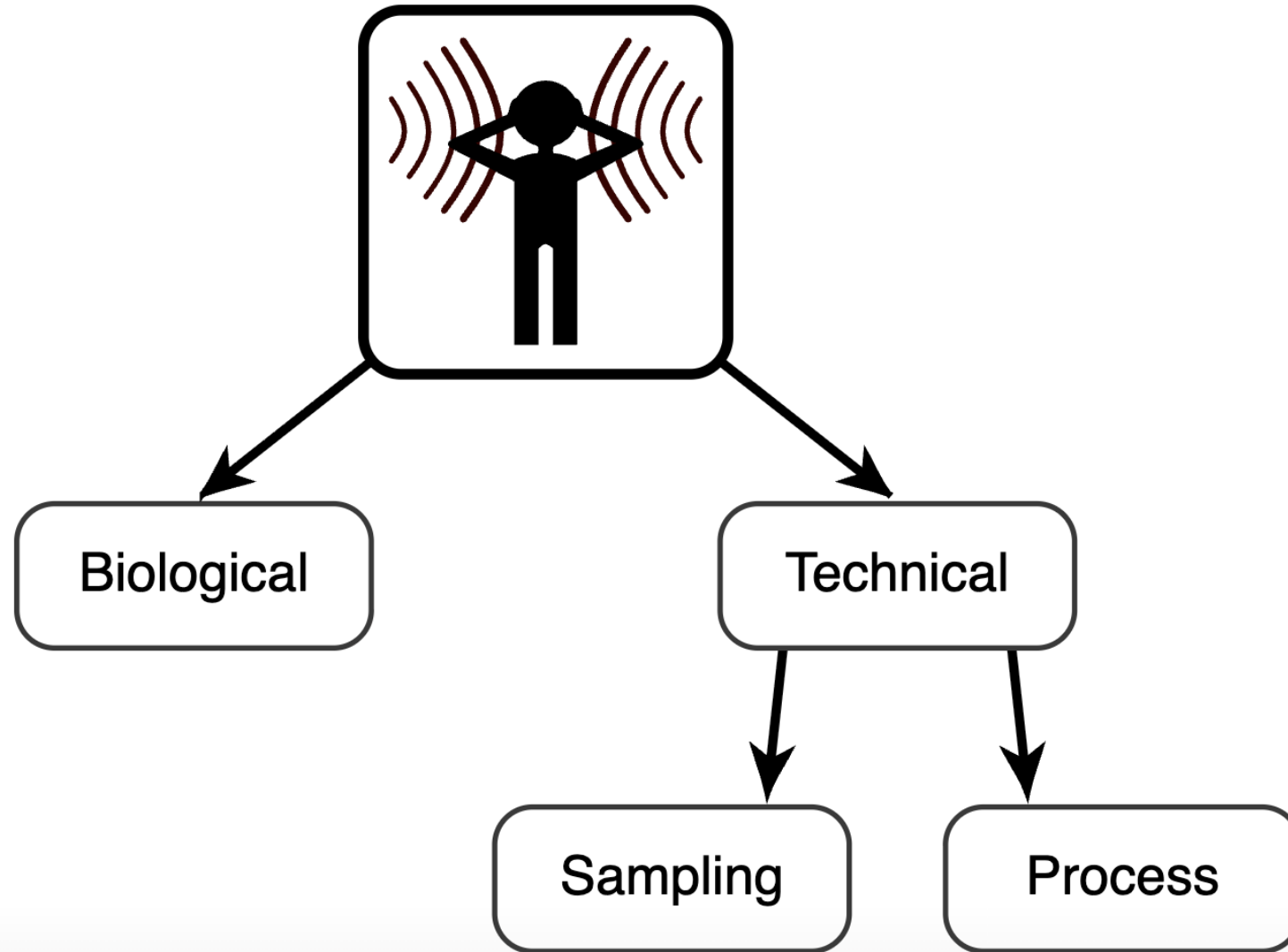


Differentially expressed genes

Gene	Treatment_1	Treatment_2	Treatment_3	Control_1	Control_2	Control_3
Gene 1	13	15	21	22	25	18
Gene 2	344	425	512	151	148	129
Gene 3	38	53	36	68	66	55
...	...	...	...	...	...	...
Gene N	2152	2382	2678	2796	2731	2745

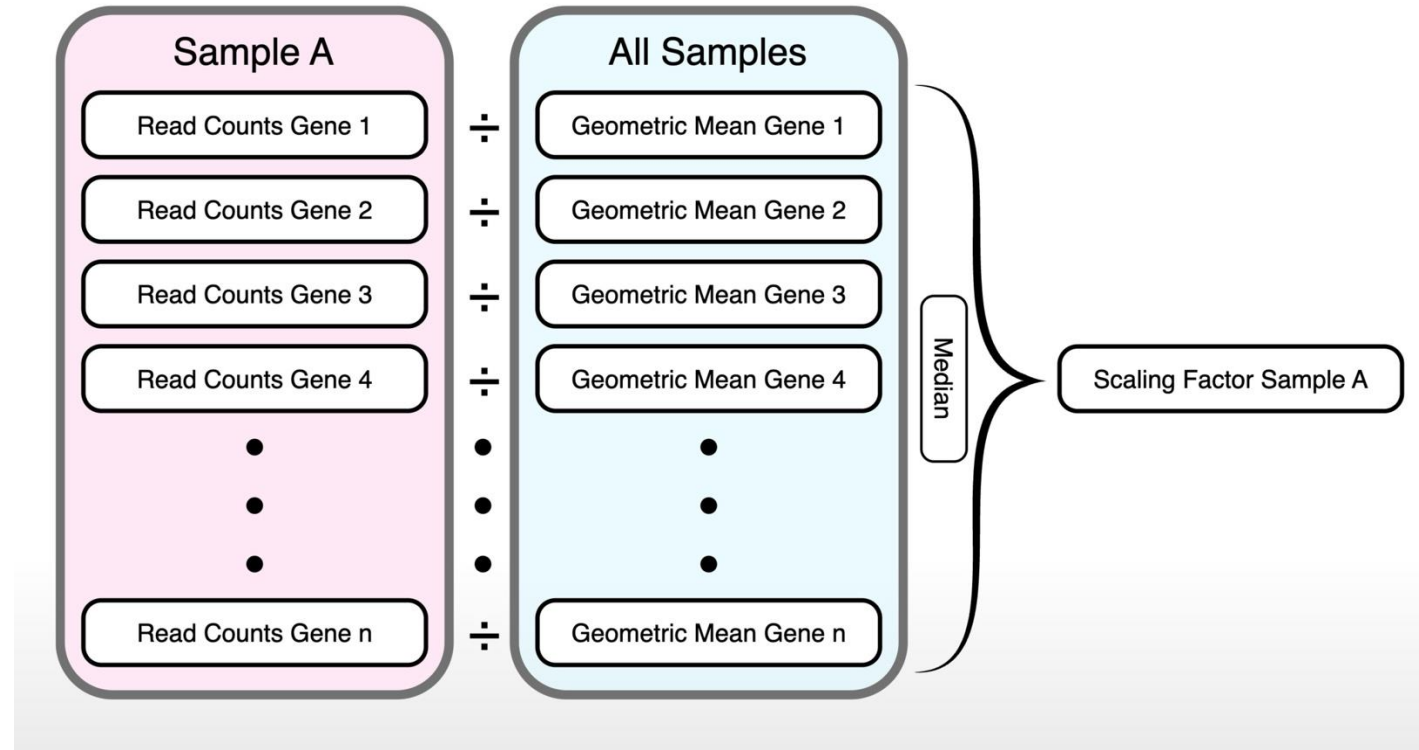
- Two groups
 - Treatment: 3 replicates
 - Control: 3 replicates
- Total of 6 samples
- Question: What genes are significantly differentially expressed genes in treatment group when compared to control group?
 - N independent statistical tests

Source of noise



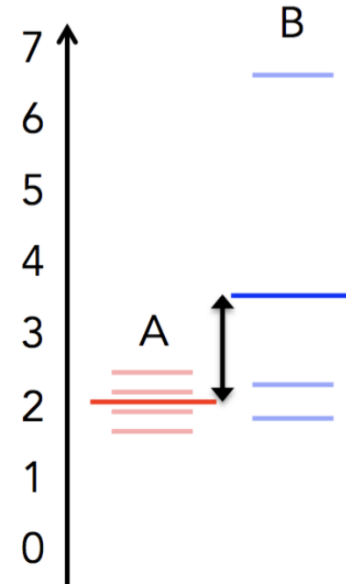
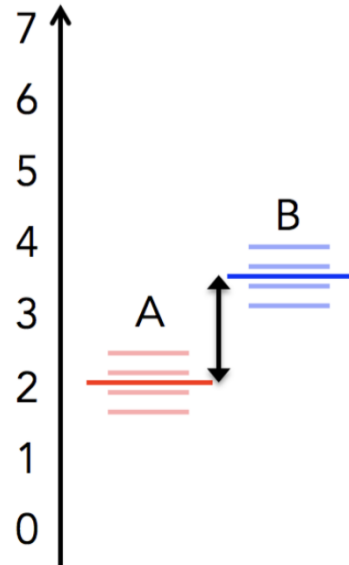
DESeq2 Normalization

- For each gene in Sample A, we take the raw read count.
- For each gene, compute the geometric mean of its read counts across all samples. This serves as a reference expression level for that gene.
- For every gene in the Sample, we compute the ratio between gene counts the its geometric mean across samples. This produces a vector of ratios.
- The median of all these gene-wise ratios is taken. This value becomes the scaling factor for Sample A.



Differential Expression

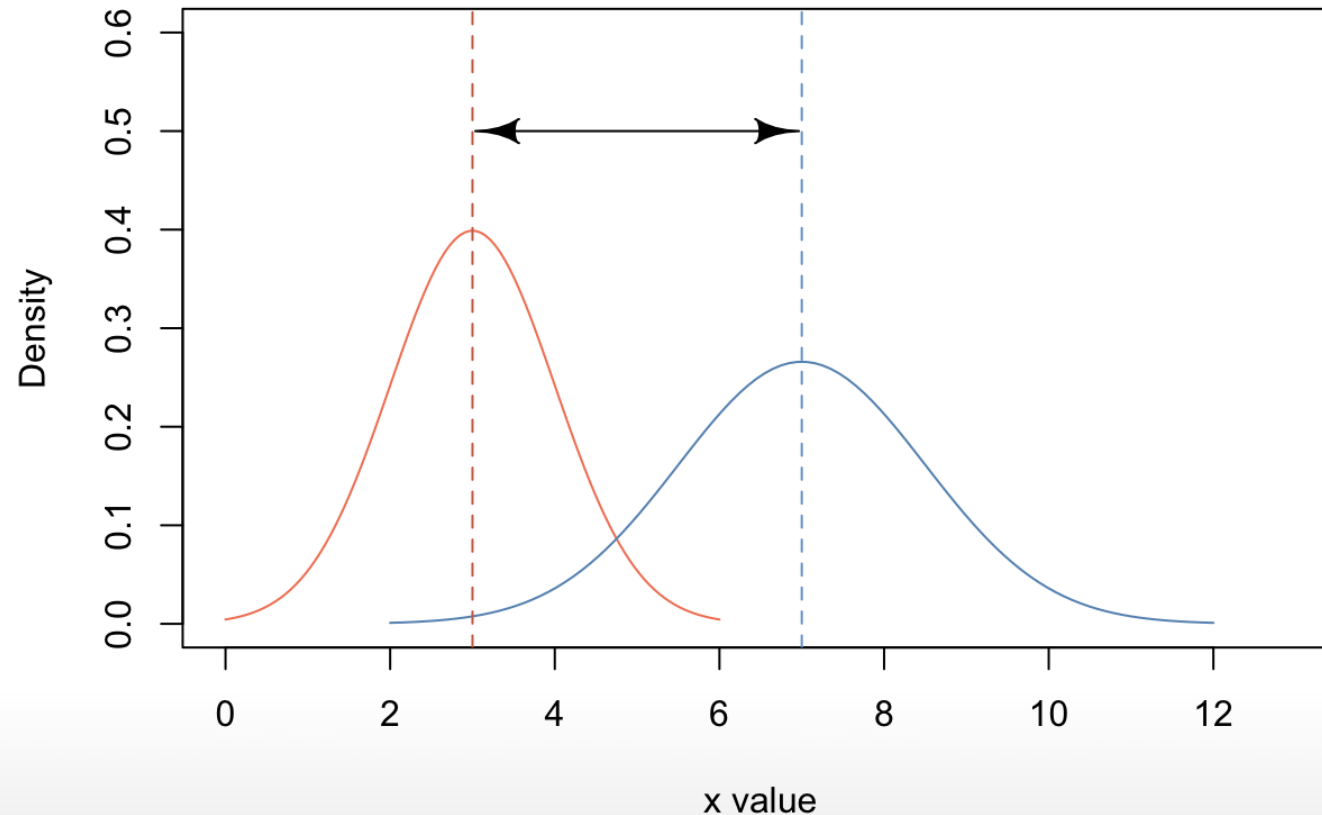
Simple difference in means



- Replication introduces variation

Differential Expression

- Normal distribution: t-test
- Suitable for microarray data but not for RNAseq data



Differential expression: NB

- a mean-dependent variance, common in raw RNA-seq count data.
- **Overdispersion**: variance grows faster than the mean (super-linear relationship).

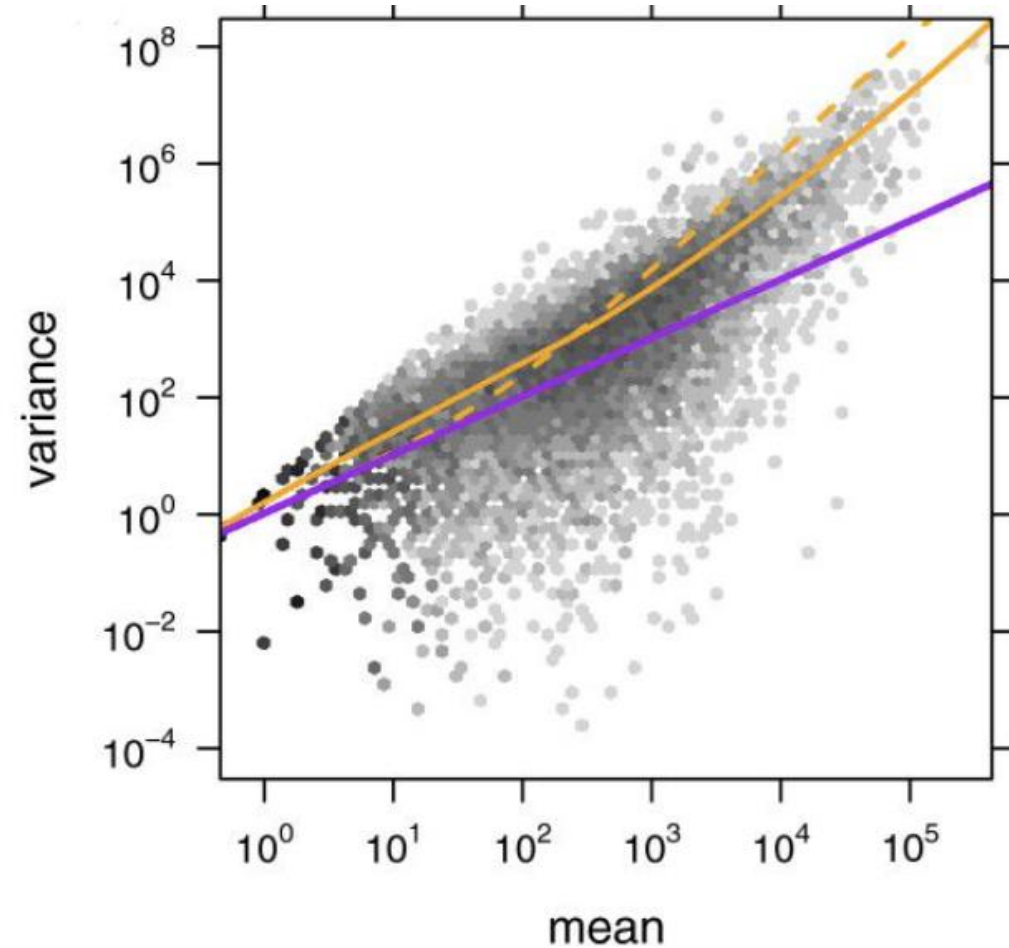
$$\text{Counts}_{ij} \sim \text{NB}(\mu_{ij}, \phi_i)$$

$$\mu_{ij} = s_j q_{ij}$$

$$\text{Var}(\text{Counts}_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$

$$\log(q_{ij}) = \beta_0 + \beta_1 \cdot \text{treatment} + \beta_2 \cdot \text{age} + \dots$$

- s is the size factor (normalization term for sequencing depth), and q_{ij} represents the true underlying expression strength of gene i in sample j .



Linear modeling in RNAseq

- Calculate size factors for each sample to account for differences in sequencing depth and transcript composition between samples
- Estimate dispersion for each gene by “borrowing” information across genes for more precise estimate when sample sizes are small (as is typical in RNA-seq experiments)
- Estimate model coefficients which are used to define test hypothesis

```
d <- DESeq(d)
```

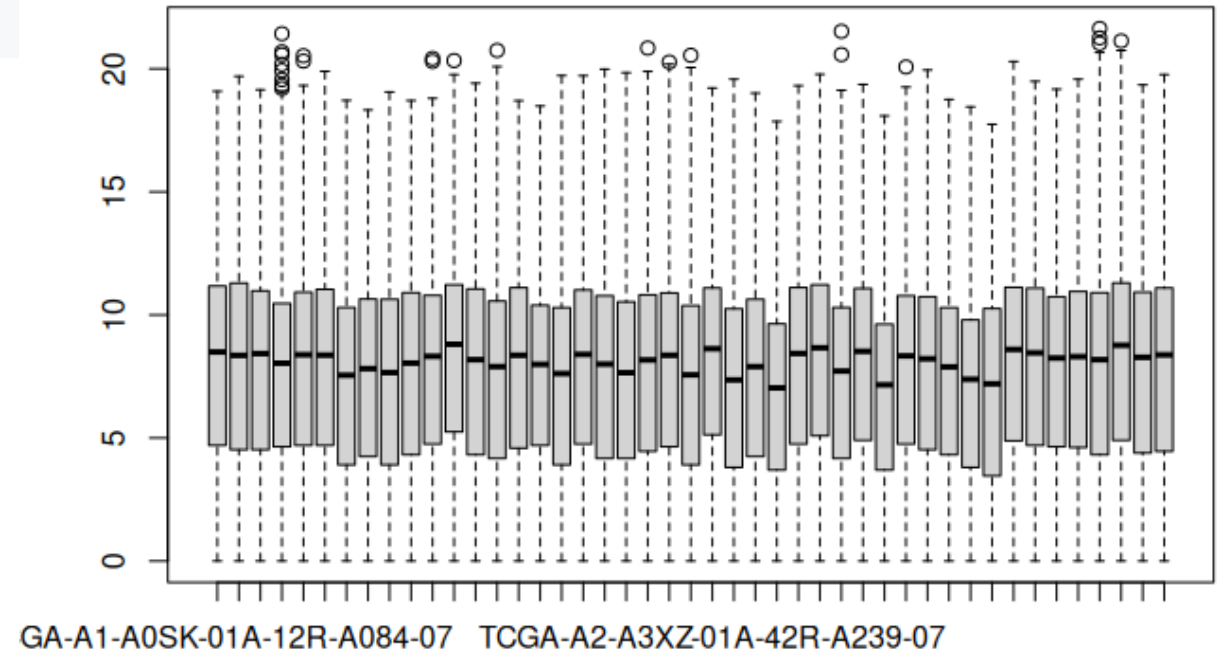
```
## estimating size factors  
  
## estimating dispersions  
  
## gene-wise dispersion estimates  
  
## mean-dispersion relationship  
  
## final dispersion estimates  
  
## fitting model and testing  
  
## -- replacing outliers and refitting for 1337 genes  
## -- DESeq argument 'minReplicatesForReplace' = 7  
## -- original counts are preserved in counts(dds)  
  
## estimating dispersions  
  
## fitting model and testing
```

Example

```
d <- DESeqDataSetFromMatrix(countData = counts,  
                             colData = coldata,  
                             design = ~ 0 + subtype)
```

```
d <- DESeq(d)
```

```
logcounts <- log2(counts(d, normalized = FALSE) + 1)  
boxplot(logcounts)
```

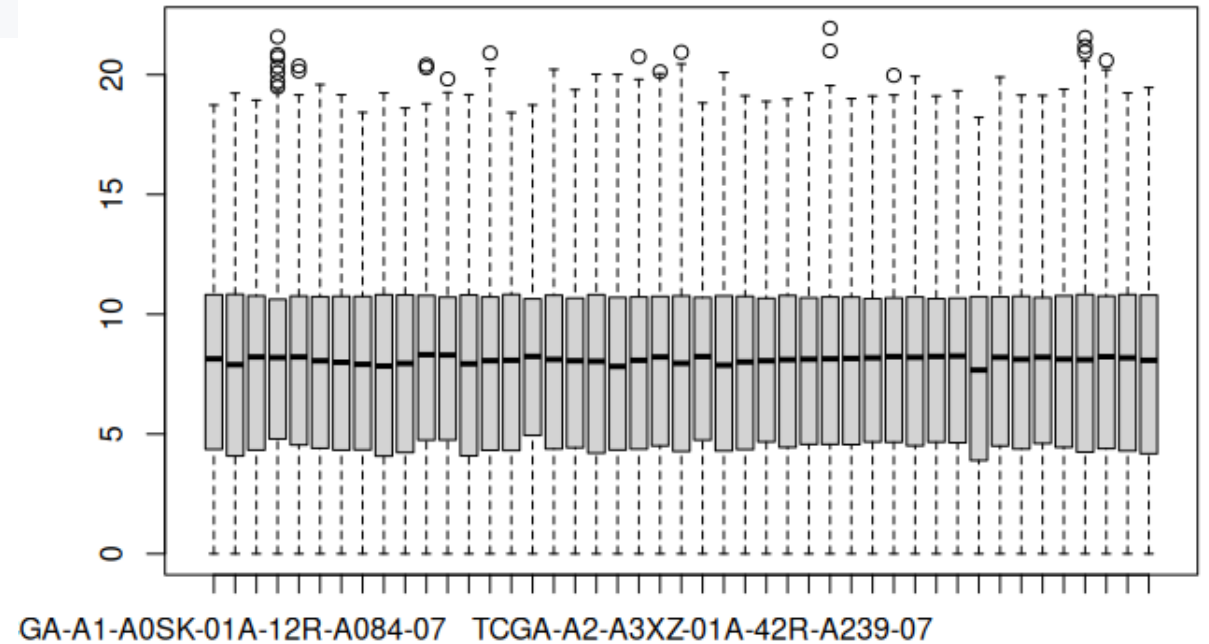


Example

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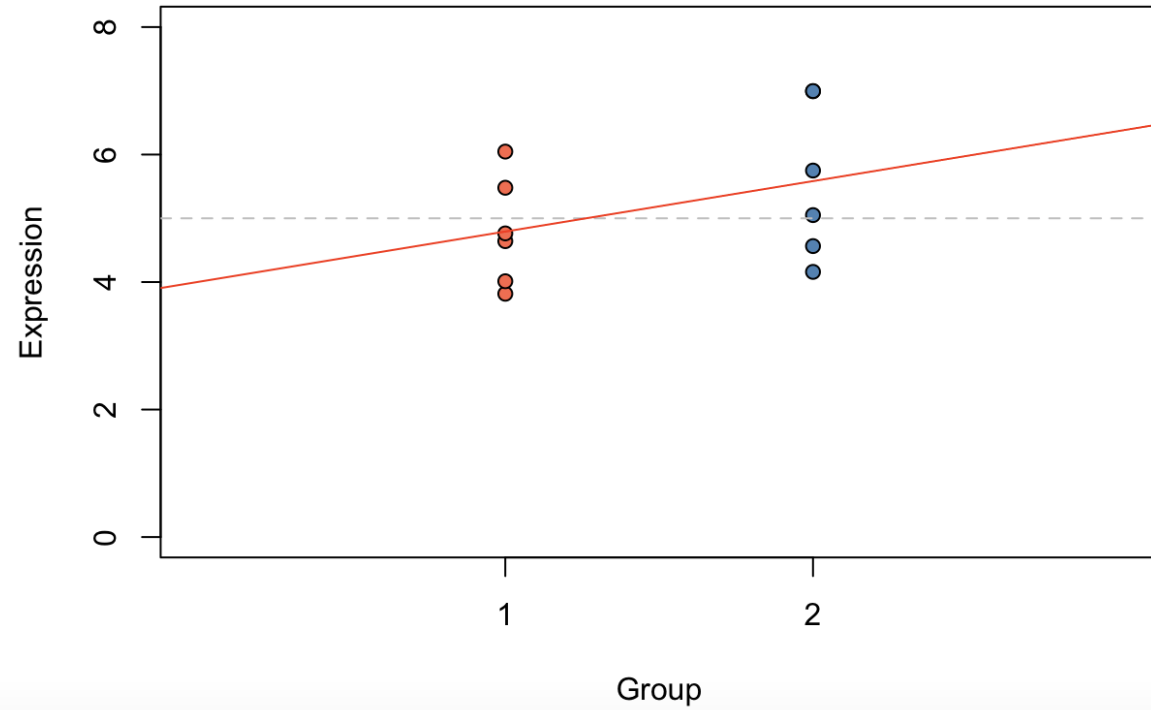
```
d <- DESeq(d)
```

```
logNormCounts <- log2(counts(d, normalized = TRUE) + 1)  
boxplot(logNormCounts)
```



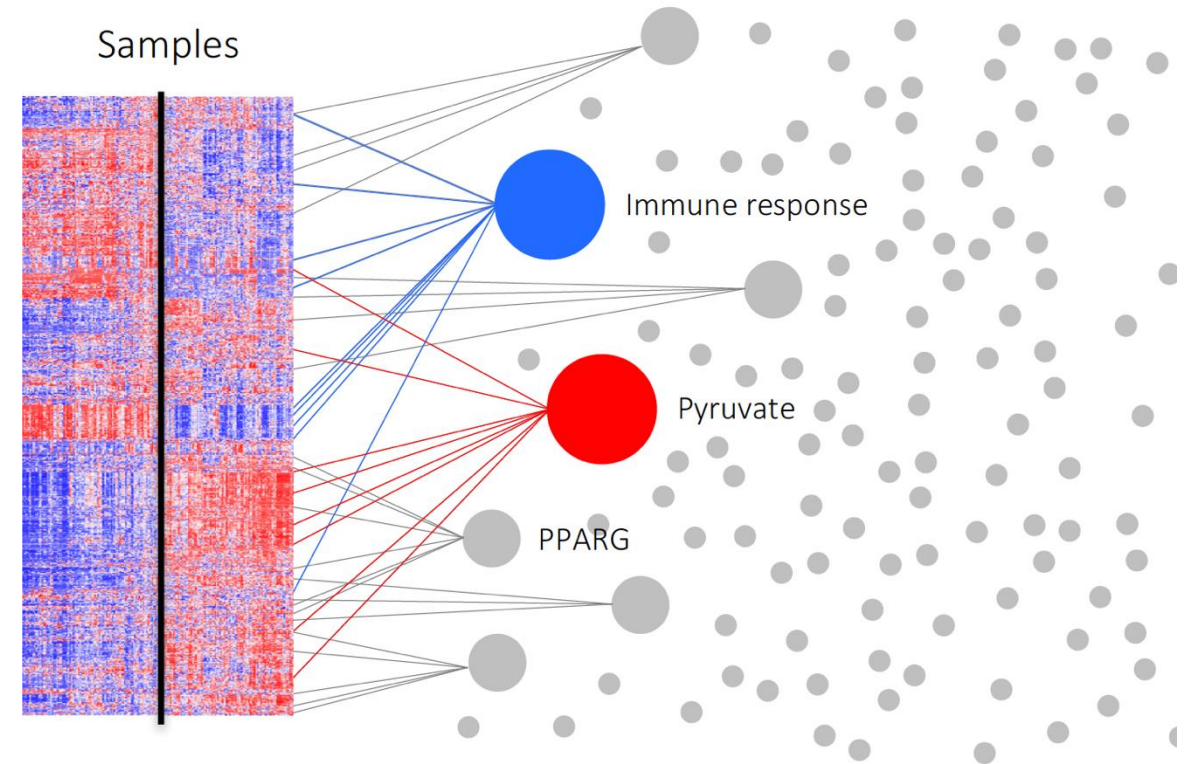
Differential expression linear model

- Calculate coefficients describing change in gene expression
- Linear model --> Generalized linear model



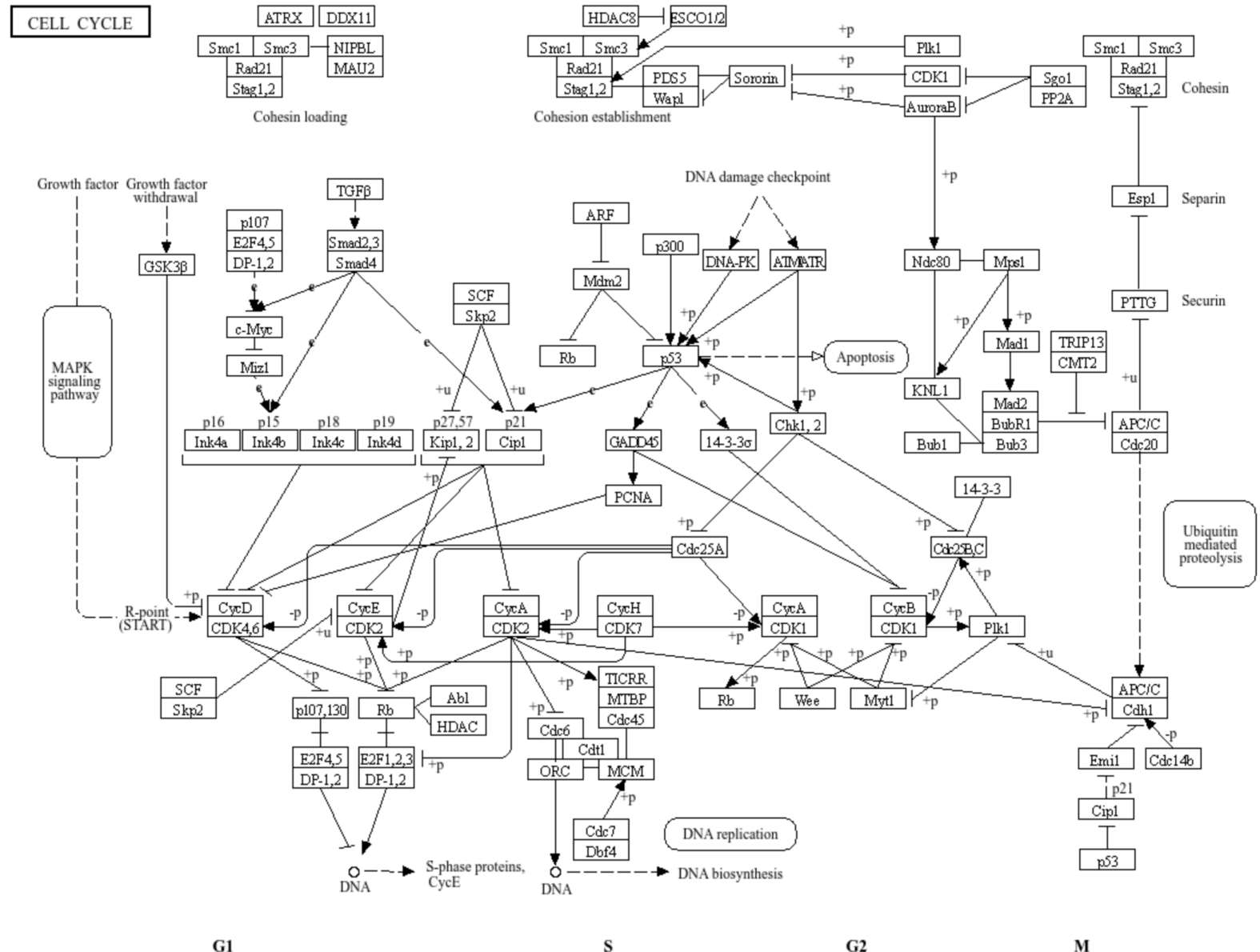
Functional annotation: gene set analysis

- Interpretation of genome-wide results
- Gene-sets are (typically) fewer than all the genes and have more descriptive names
- Difficult to manage a long list of significant genes
- Detect patterns that would be difficult to discern simply by manually going through e.g. the list of differentially expressed genes
- Integrates external information into the analysis
- Less prone to false-positives on the gene-level
- Top genes might not be the interesting ones, several coordinated smaller changes

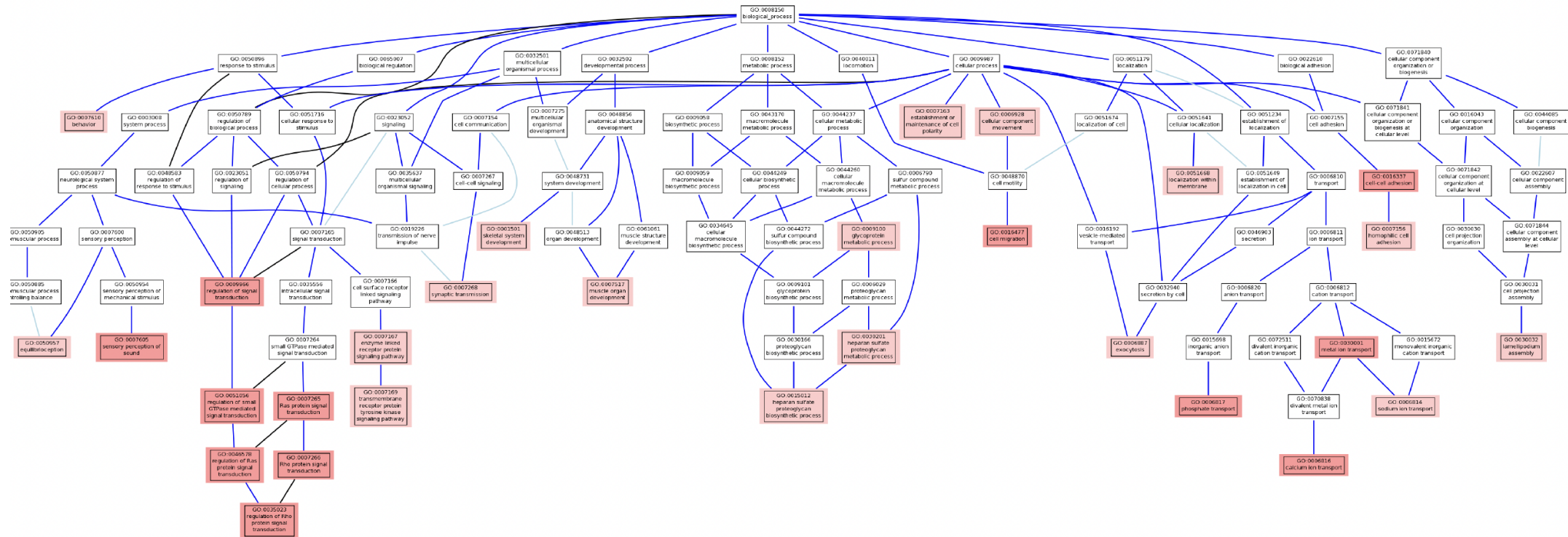


What is a pathway?

- Metabolic pathways are a series of chemical reactions occurring within a cell.
- A pathway describes a biological function/process specifically



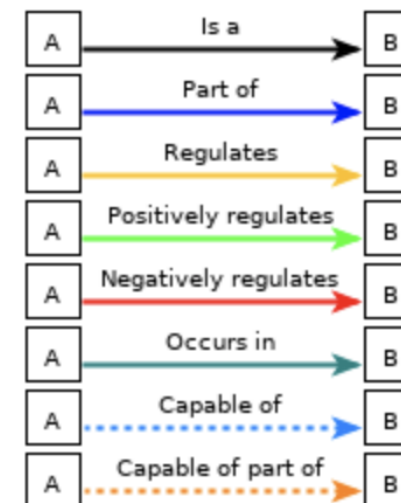
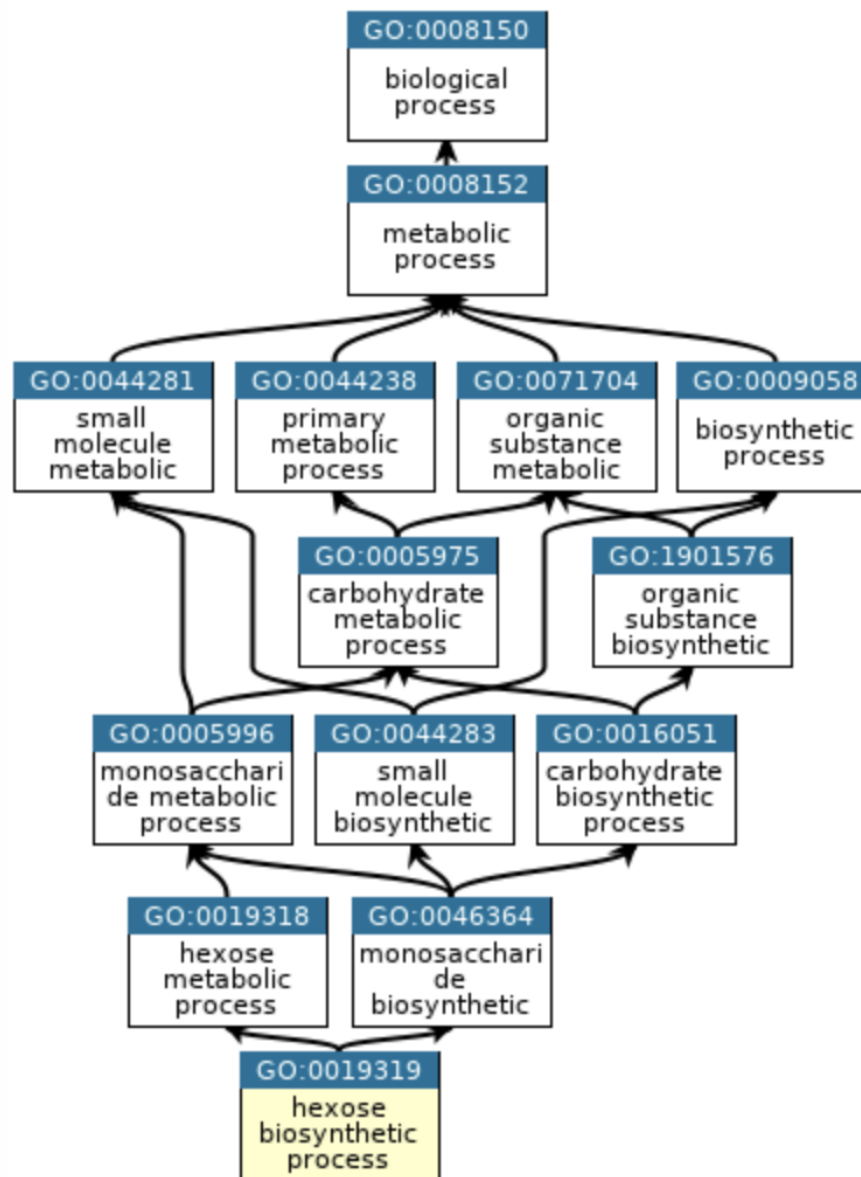
Example of gene sets: Gene Ontology



- Hierarchical graph with three categories:
- **Biological process, Molecular function, Cellular compartment**
- Terms get more and more detailed moving down the hierarchy
- Genes can belong to multiple GO terms

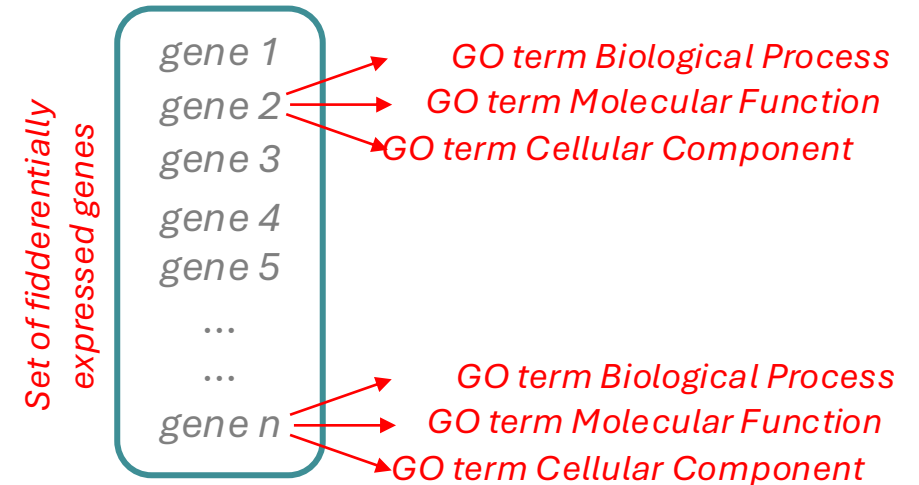
GO

- Three controlled vocabularies or ontologies of terms (called GO terms) have been developed that, for each gene or its product, describe:
- **Molecular Function.**
This refers to the molecular activity performed by the gene product, e.g., *catalytic activity, transporter activity, Toll-like receptor binding*, etc.
- **Biological Process.**
The pathway or process in which the gene product is involved. A biological process is defined as “a mechanism controlled by multiple gene products with different molecular functions that leads to a specific outcome.” A biological process is carried out and completed through an ordered set of molecular functions and often involves a chemical or physical transformation, e.g., *DNA repair, signal transduction, cellular division*.
- **Cellular Component.**
The cellular or extracellular compartment in which a gene product’s activity takes place, e.g., *mitochondrion, ribosome*, etc.



How we use the gene ontology for gene set enrichment analysis

- To understand the role (in functional terms) of a set of genes (e.g., those that are differentially expressed) within the three main GO categories.
- **Q:** Is there a functional category (GO term) that is particularly represented (enriched) in the gene set?
- **A:** It is possible that a certain fraction of genes in the gene set are associated with a specific biological process or molecular function that could explain the phenotypic differences between the compared conditions.
- Note: Observing that a certain fraction of genes is associated with a given process is not sufficient → it is necessary to estimate the statistical significance to verify whether a given annotation occurs more (or less) frequently in the gene set of interest than in a random sampling of genes.



Overrepresentation analysis

- **Example.** Suppose we analyze the expression profile (6,448 genes) of a total number of yeast samples. Among these, we identify 100 differentially expressed genes (DEGs). We notice that 20 of these genes are annotated with the GO term “DNA replication.”
- **Q:** Is the abundance of genes annotated with “DNA replication” significant, or is it a coincidence due to chance?

	DEGs	NON DEGs
DNA replication	20	94
NON DNA replication	80	6348

$N = 6,448$ (total number of genes in yeast)

$M = 114$ (total number of genes annotated with “DNA replication”)

$n = 100$ (number of DEGs / number of randomly selected genes)

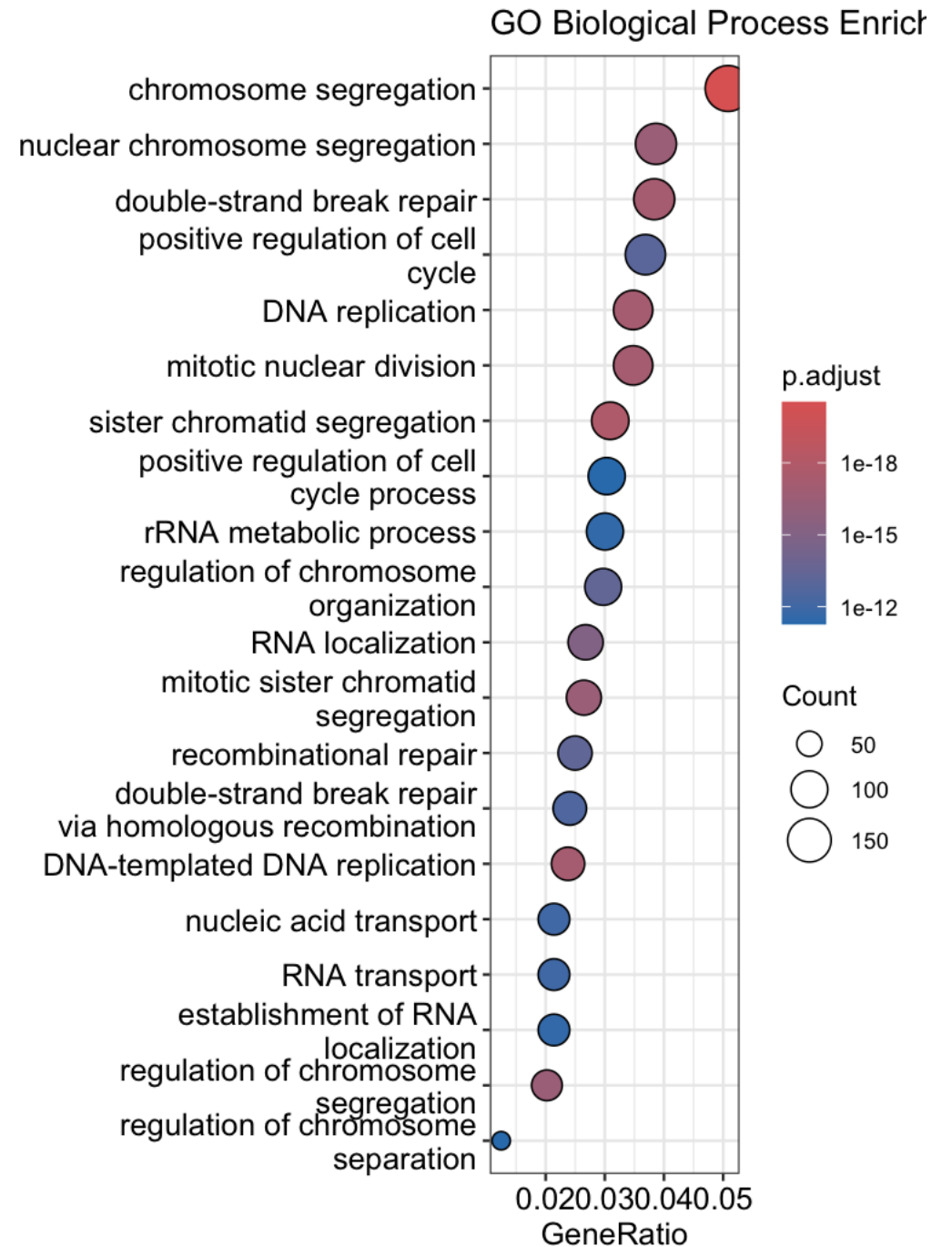
$m = 20$ (number of DEGs annotated with “DNA replication”)

$$p(N, M; n, m) = \frac{\text{observed cases}}{\text{all possible cases}} = \frac{\binom{M}{m} \binom{N-M}{n-m}}{\binom{N}{n}} = 1.5E^{-17}$$

Example

```
ego <- enrichGO(  
  gene      = res_basal_sig$ENSEMBL, #the list of DE genes  
  OrgDb     = org.Hs.eg.db, # the organisms (background)  
  keyType   = "ENSEMBL", #annotation of the list  
  ont       = "BP", # The ontology we want to use ("MF" or "CC" are also available)  
  pAdjustMethod = "BH", # Multiple test adjustment method  
  pvalueCutoff = 0.01, # Cutoff on the p-value  
  qvalueCutoff = 0.05 # Cutoff on the q-value  
)
```

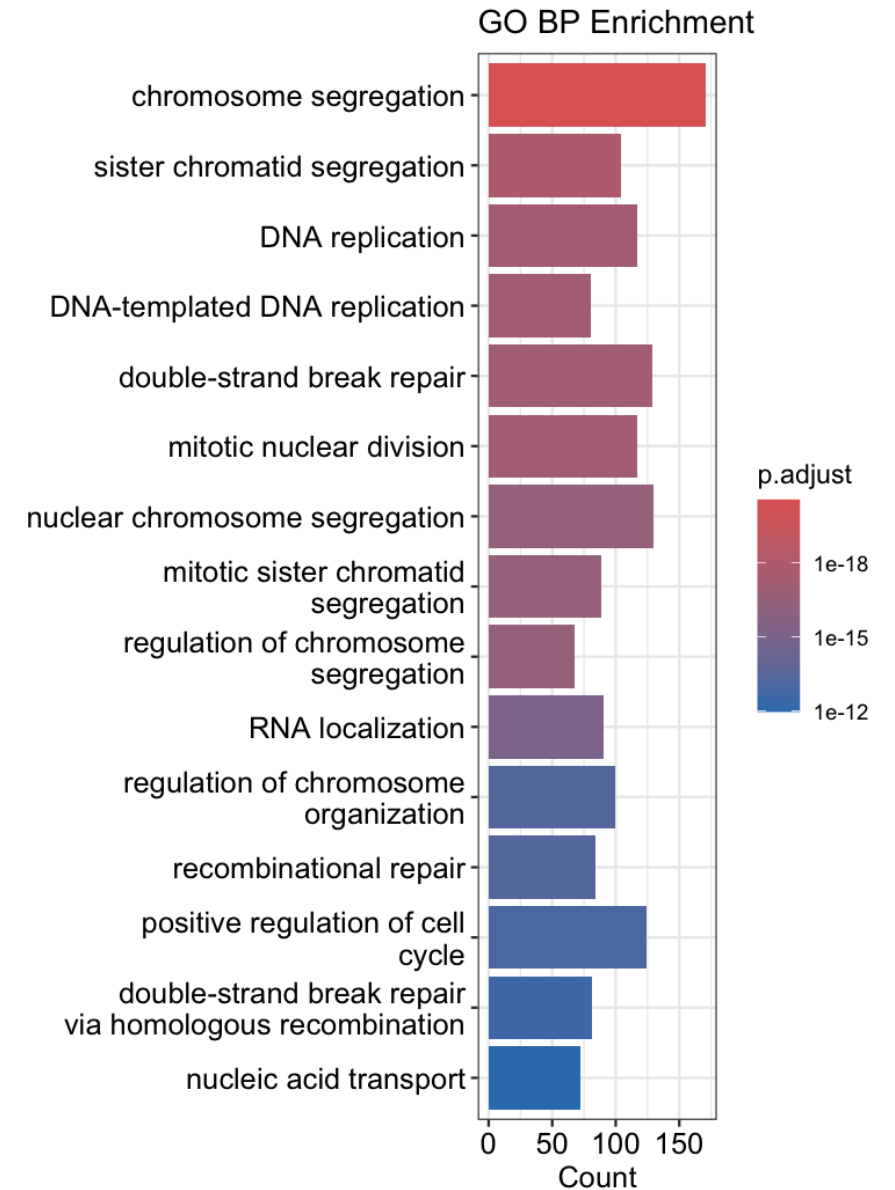
```
dotplot(ego, showCategory = 20) + ggtitle("GO Biological Process Enrichment of Basal tumors")
```



Example

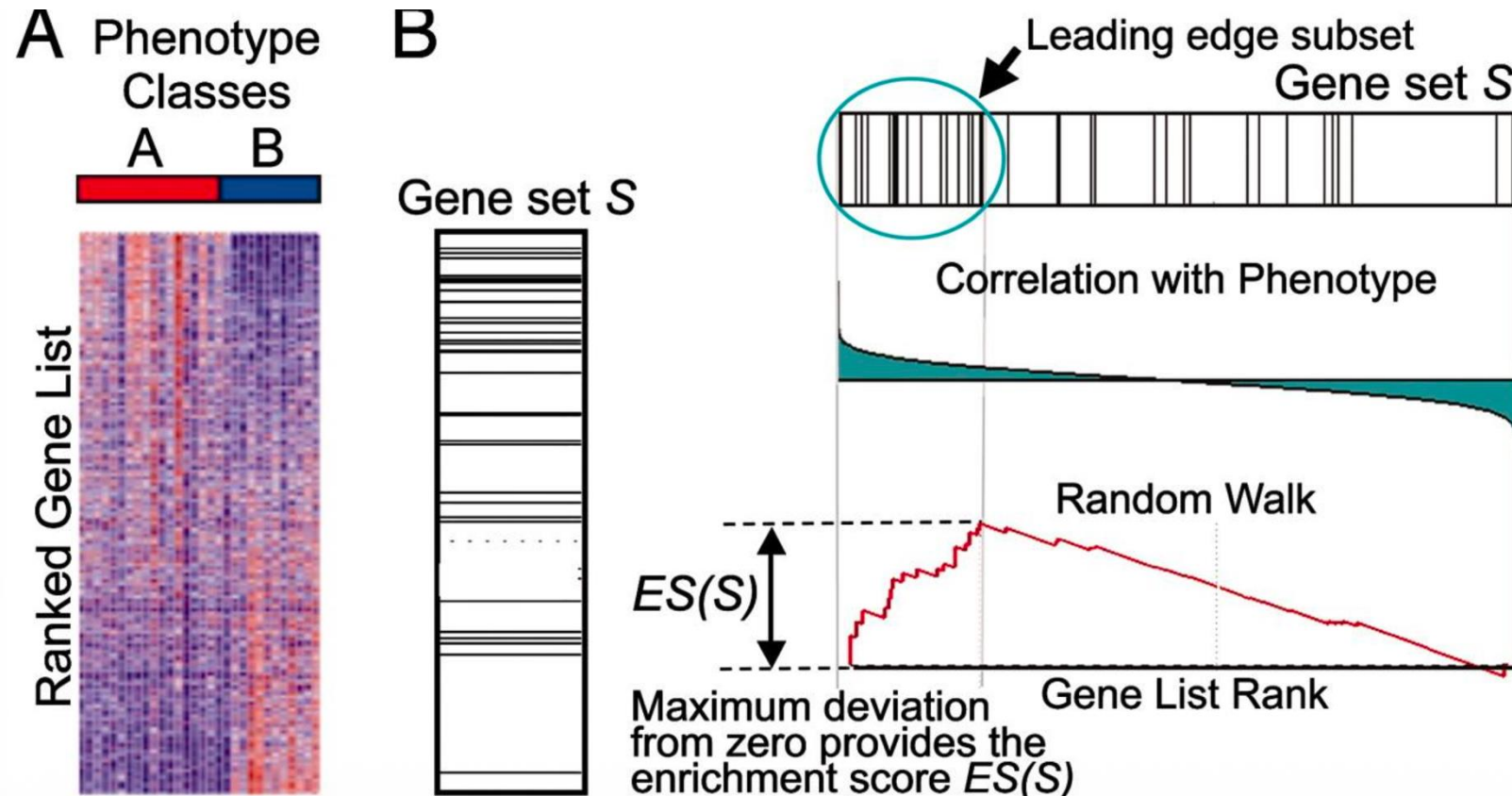
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  pvalueCutoff = 0.01, # Cutoff on the p-value  
  qvalueCutoff = 0.05 # Cutoff on the q-value  
)
```

```
barplot(ego, showCategory = 15, title = "GO BP Enrichment", sortBy = "pvalue")
```



Gene Set Enrichment Analysis (GSEA)

- Over-representation analysis requires a threshold to select the DEGs and compute the contingency table and apply Fisher test
- GSEA instead uses the whole list of gene.



Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles

Aravind Subramanian^{a,b}, Pablo Tamayo^{a,b}, Vamsi K. Mootha^{a,c}, Sayan Mukherjee^d, Benjamin L. Ebert^{a,e}, Michael A. Gillette^{a,f}, Amanda Paulovich^g, Scott L. Pomeroy^h, Todd R. Golub^{a,i}, Eric S. Lander^{a,k,l,m}, and Jill P. Mesirov^{a,h}

Gene Set Enrichment Analysis (GSEA)

- Given: a set S of genes and a list L of genes ranked by correlation (or other metric) between two conditions/classes/phenotypes
- Question: is S randomly distributed in L or is S focused at one of the ends?

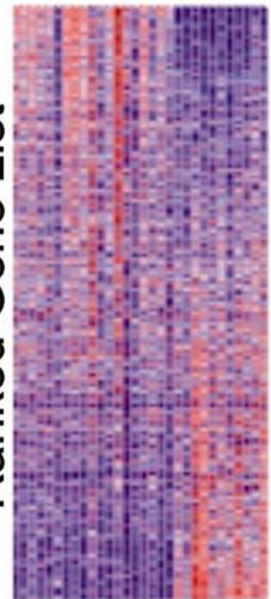
A Phenotype

Classes

A B

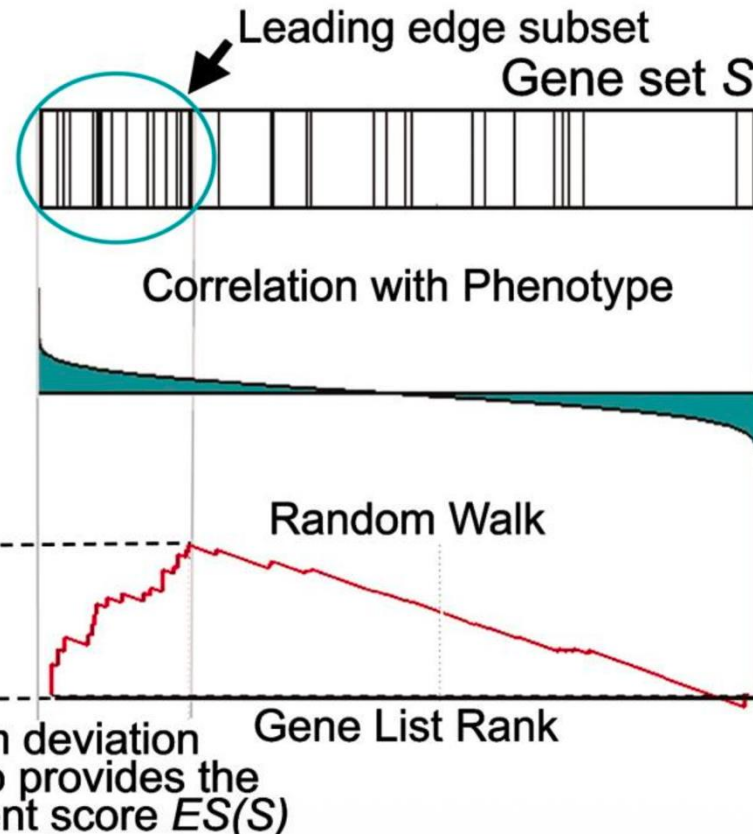
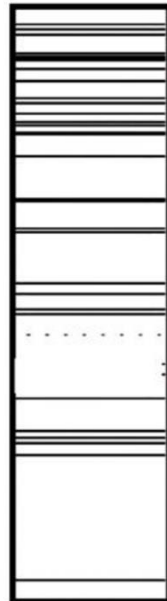


Ranked Gene List



B

Gene set S



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GSEA details

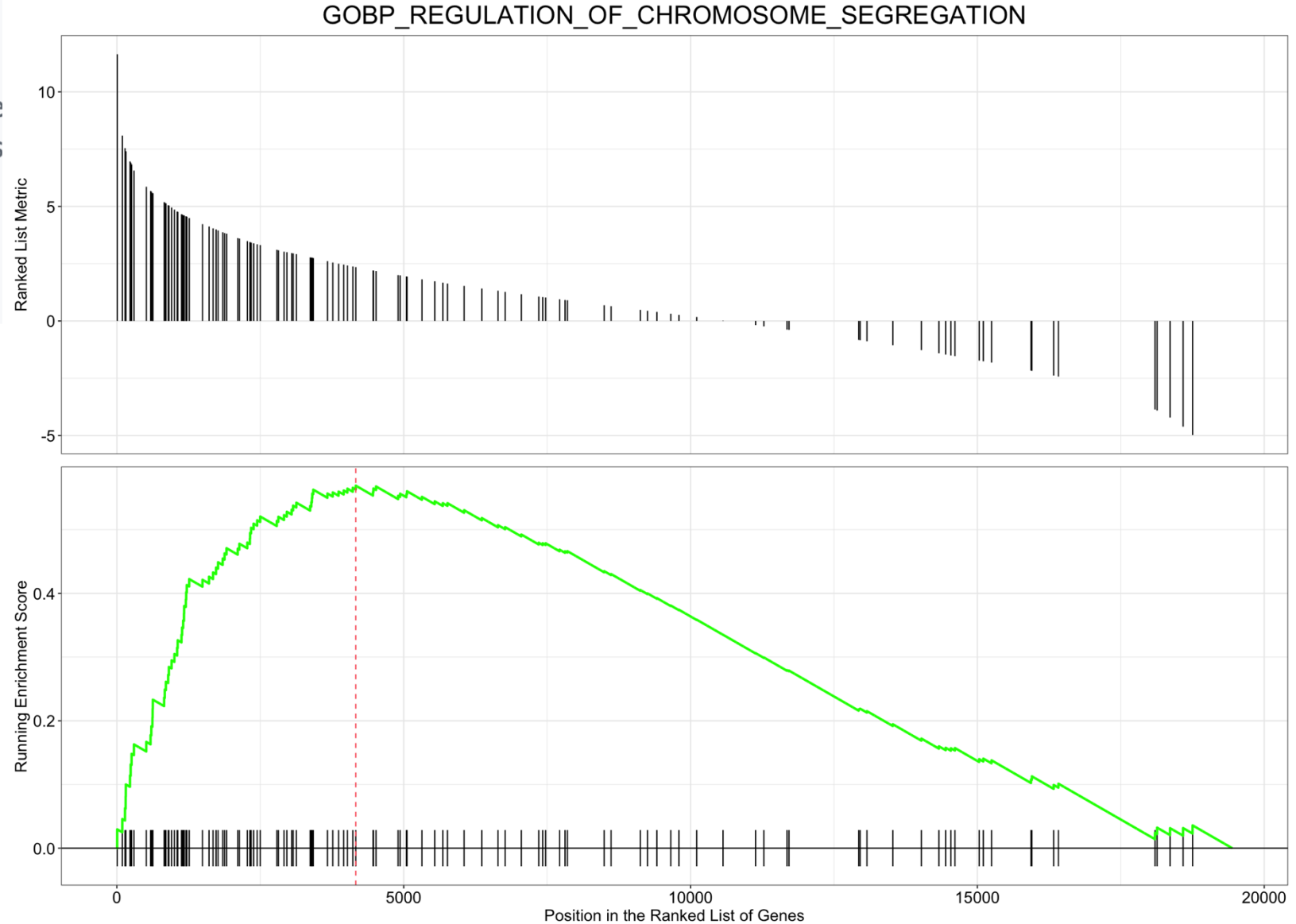
$$P_{\text{hit}}(S, i) = \sum_{g_j \in S, j \leq i} \frac{|r_j|^p}{N_R}, \quad \text{where} \quad N_R = \sum_{g_j \in S} |r_j|^p$$

$$P_{\text{miss}}(S, i) = \sum_{g_j \notin S, j \leq i} \frac{1}{(N - N_H)}.$$

- Find the largest (inc. negative) value for $P_{\text{hit}} - P_{\text{miss}}$
- When p is 0, this is the fraction of genes in S versus not in S up until point i
- This case happens to correspond to the Kolmogorov-Smirnov statistic

GSEA Example

```
#perform the GSEA enrichment analysis
gsea_go <- GSEA(
  geneList      = geneList, #sorted gene
  TERM2GENE     = term2gene, #the gene s
  pvalueCutoff  = 0.05,
  verbose      = FALSE
)
```



```
gseaplot(gsea_go, geneSetID = 1, title = gsea_go$Description[1])
```